2. *Hadoop MapReduce:* This works as a parallel framework for scheduling and processing the data.
3. *Hadoop YARN:* This is an acronym for Yet Another Resource Navigator. It is an improved version of MapReduce and is used for processes running over Hadoop.
4. *Hadoop Distributed File System – HDFS:* This stores data and maintains records over various machines or clusters. It also allows the data to be stored in an accessible format.

HDFS sends data to the server once and uses it as many times as it wants. When a query is raised, NameNode manages all the DataNode slave nodes that serve the given query. Hadoop MapReduce performs all the jobs assigned sequentially. Instead of MapReduce, Pig Hadoop and Hive Hadoop are used for better performances.

Other packages that can support Hadoop are listed below.

- *Apache Oozie:* A scheduling system that manages processes taking place in Hadoop
- *Apache Pig:* A platform to run programs made on Hadoop
- *Cloudera Impala:* A processing database for Hadoop. Originally it was created by the software organisation Cloudera, but was later released as open source software
- *Apache HBase:* A non-relational database for Hadoop
- *Apache Phoenix:* A relational database based on Apache HBase
- *Apache Hive:* A data warehouse used for summarisation, querying and the analysis of data
- *Apache Sqoop:* Is used to store data between Hadoop and structured data sources
- *Apache Flume:* A tool used to move data to HDFS
- *Cassandra:* A scalable multi-database system

The importance of Hadoop

Hadoop is capable of storing and processing large amounts of data of various kinds. There is no need to preprocess the data before storing it. Hadoop is highly scalable as it can store and distribute large data sets over several machines running in parallel. This framework is free and uses cost-efficient methods.

Hadoop is used for:

- Machine learning
- Processing of text documents
- Image processing
- Processing of XML messages
- Web crawling
- Data analysis
- Analysis in the marketing field
- Study of statistical data

Challenges when using Hadoop

Hadoop does not provide easy tools for removing noise from the data; hence, maintaining that data is a challenge. It has many data security issues like encryption problems. Streaming jobs and batch jobs are not performed efficiently. MapReduce programming is inefficient for jobs involving highly analytical skills. It is a distributed system with low level APIs. Some APIs are not useful to developers.

But there are benefits too. Hadoop has many useful functions like data warehousing, fraud detection and marketing campaign analysis. These are helpful to get useful information from the collected data. Hadoop has the ability to duplicate data automatically. So multiple copies of data are used as a backup to prevent loss of data.

Frameworks similar to Hadoop

Any discussion on Big Data is never complete without a mention of Hadoop. But like with other technologies, a variety of frameworks that are similar to Hadoop have been developed. Other frameworks used widely are Ceph, Apache Storm, Apache Spark, DataTorrentRTS, Google BiqQuery, Samza, Flink and HydraDataTorrentRTS.

MapReduce requires a lot of time to perform assigned tasks. Spark can fix this issue by doing in-memory processing of data. Flink is another framework that works faster than Hadoop and Spark. Hadoop is not efficient for real-time processing of data. Apache Spark uses stream processing of data where continuous input and output of data happens. Apache Flink also provides single runtime for the streaming of data and batch processing.

However, Hadoop is the preferred platform for Big Data analytics because of its scalability, low cost and flexibility. It offers an array of tools that data scientists need. Apache Hadoop with YARN transforms a large set of raw data into a feature matrix which is easily consumed. Hadoop makes machine learning algorithms easier. END

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By: Jameer Babu

The author is a FOSS enthusiast and is interested in competitive programming and problem solving. He can be contacted at *jameer.jb@gmail.com*.